

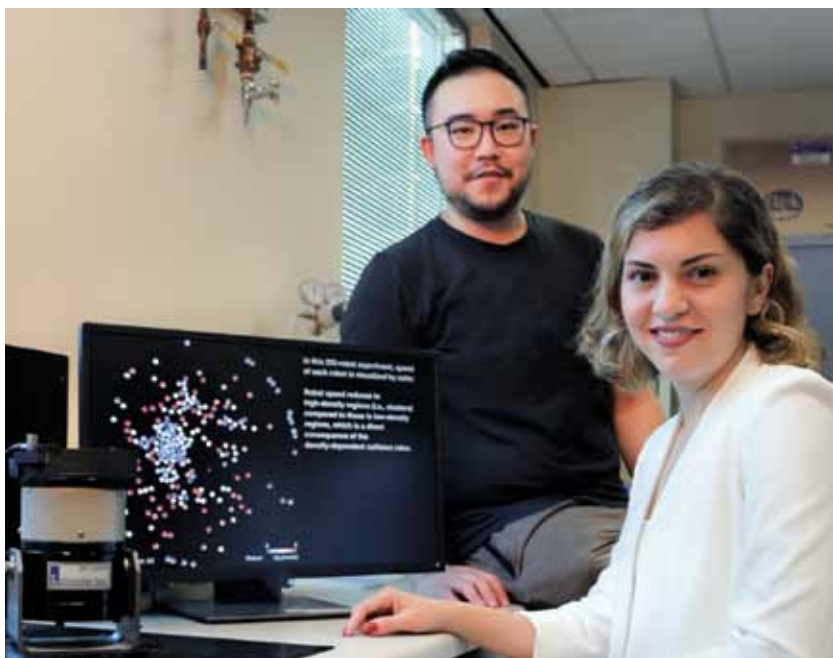


RESEARCH

& INNOVATION UPDATES

Swarms of microbots controlled using vibrations

Researchers at Georgia Institute of Technology have developed a system to control swarms of 300 3mm micro-bristle robots known as microbots. They use vibrations to control the microbots' motility characteristics—the ability of an 'organism' to move independently. This concept is inspired from thermodynamics, when an agitated material can change phases from solid to gas to liquid. Researchers manipulated the level of vibration to influence the microbots to form clusters or disperse to create good spatial coverage. They have also developed computational models and a live tracking system for the 300-robot swarm using computer vision to better understand these phase separations. This enabled the researchers to analyse microrobots' behaviour and motion data that give rise to the swarm's characteristics.



Zhijian Hao (left) and Azadeh Ansari in their lab (Credit: <https://techxplore.com>)

Mushroom skin replaces plastic as chip base



A team of researchers at Johannes Kepler University has developed a way to narrow down the use of plastic in electronic devices that causes environmental hazards. Researchers came across *Ganoderma lucidum*, a type of mushroom that grows a skin to cover its mycelium—its root-like part. They recorded that this skin was flexible and provided good insulation as well as it could withstand high temperatures. These properties of the mushroom skin can make it a very good chip substrate. The team developed a means for depositing metal electronic circuitry components onto the skin using physical vapour deposition, which was followed up with an ablated laser. Results showed that the skin worked nearly as well as the traditional plastic substrates and that it could withstand being bent repeatedly—they found no breakage after 2,000 bends.

Fungal-grown mycelium skins are introduced as bioderived, degradable and sustainable substrate materials for electronic sensor skins and batteries (Credit: <https://techxplore.com>)